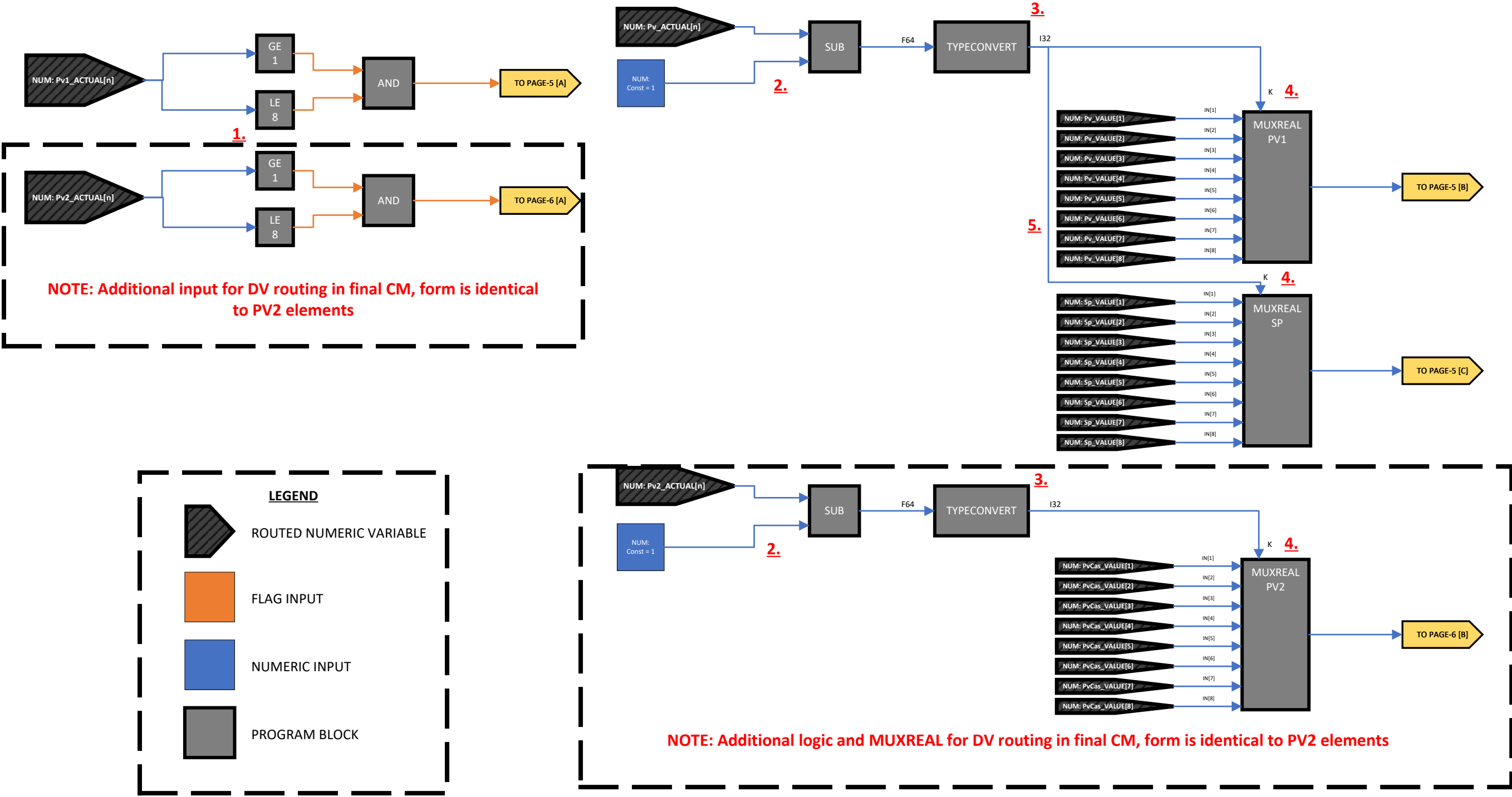
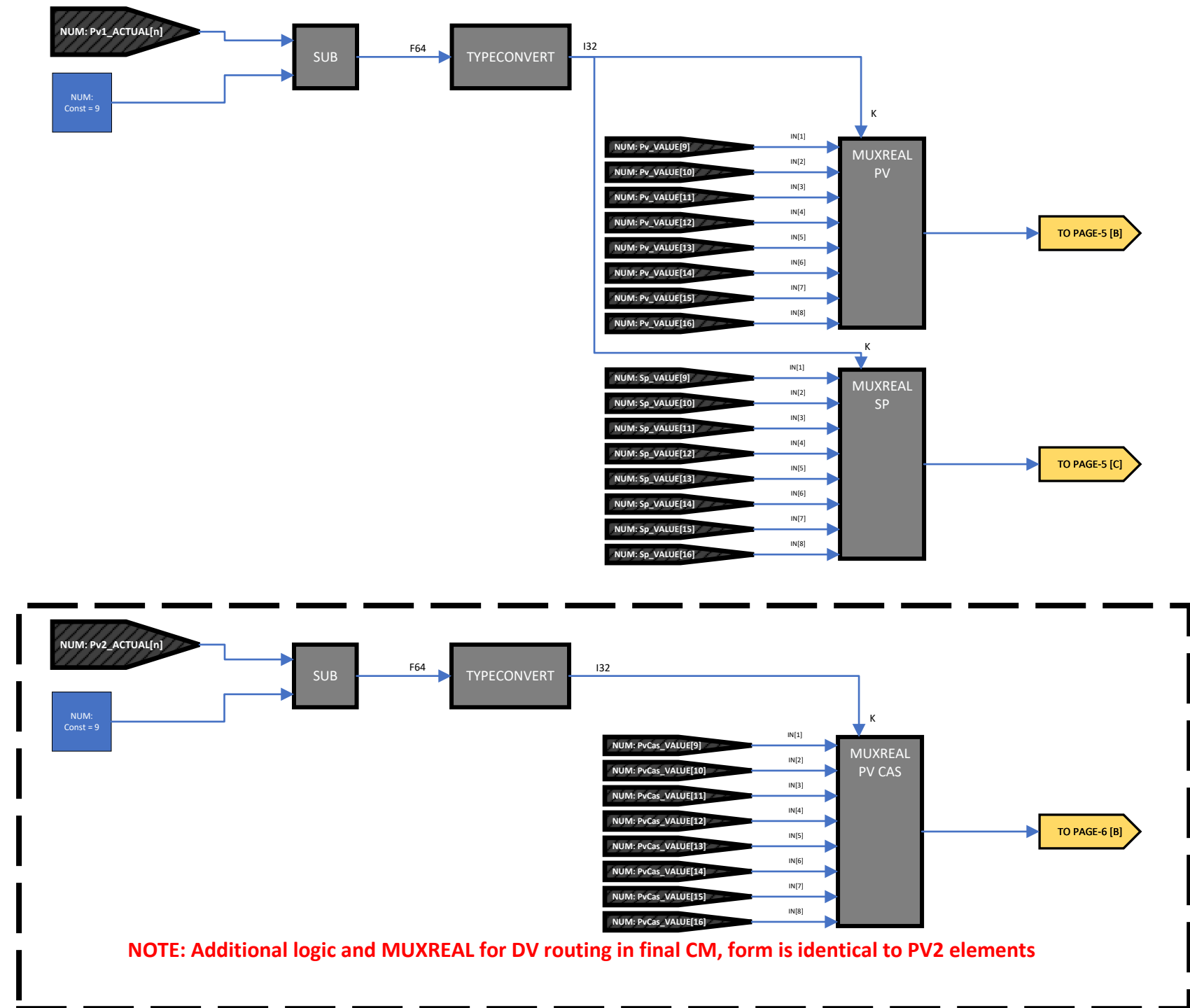




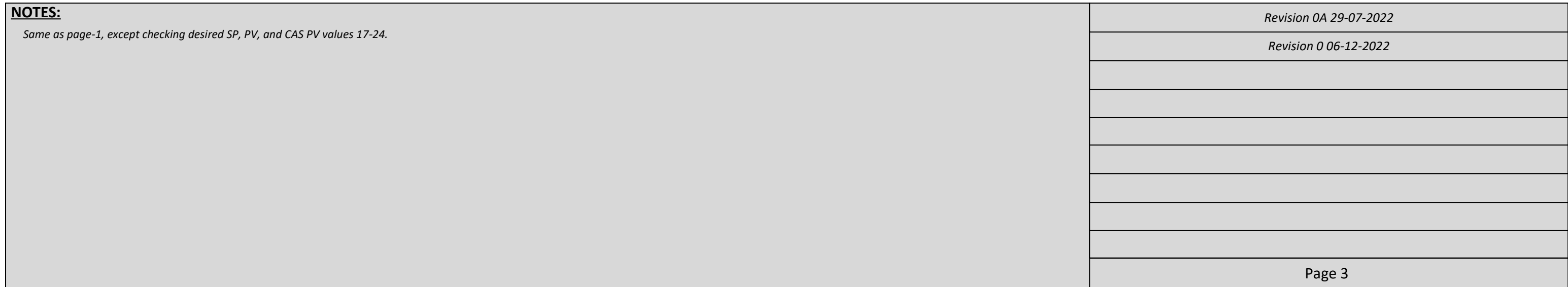
- NOTES:**
- Pv_ACTUAL[n] and Pv2_ACTUAL[n] are routed from CM "SpPvPvcas_Array". These variables establish the selected main loop process variable, and CASCADE loop process variable for the given manipulated variable "n".
 - The MUXREAL block is limited to 8 inputs, as there can be up to 32 selectable main loop, and cascade loop PVs "greater than/equal to", "less than/equal to" blocks are used to activate the outputs associated with the selections 1-8, 9-16, 17-24, 25-32, respectively. The selection itself is made on pages 5 and 6.
 - MUXREAL routes an input 1-8 to the output according to the input K, where K=0 corresponds to IN[1], K=1 corresponds to IN[2], ..., K=7 corresponds to IN[8]. As a result we need to map the input selection Pv_ACTUAL[n] or PvCas_ACTUAL[n] to this range. For this page we are checking for selection=1-8, thus we subtract 1 to map for the next page we are checking for selection 9-16, thus we subtract 9 to map, etc., for pages 3 and 4.
 - Conversion from 64-bit float to 32-bit int required for MUXREAL.
 - Input K is clamped to 0-7, thus, for input <0, K=0 fed into MUX, and for input >7, K=7 fed into MUX. Thus "greater than/equal to", "less than/equal to" flag also required for correct routing as is done on pages 5 and 6.
 - Each PV has a corresponding SP, making a unique SP, PV pair. Thus, when routing PV, the corresponding SP must also be routed.

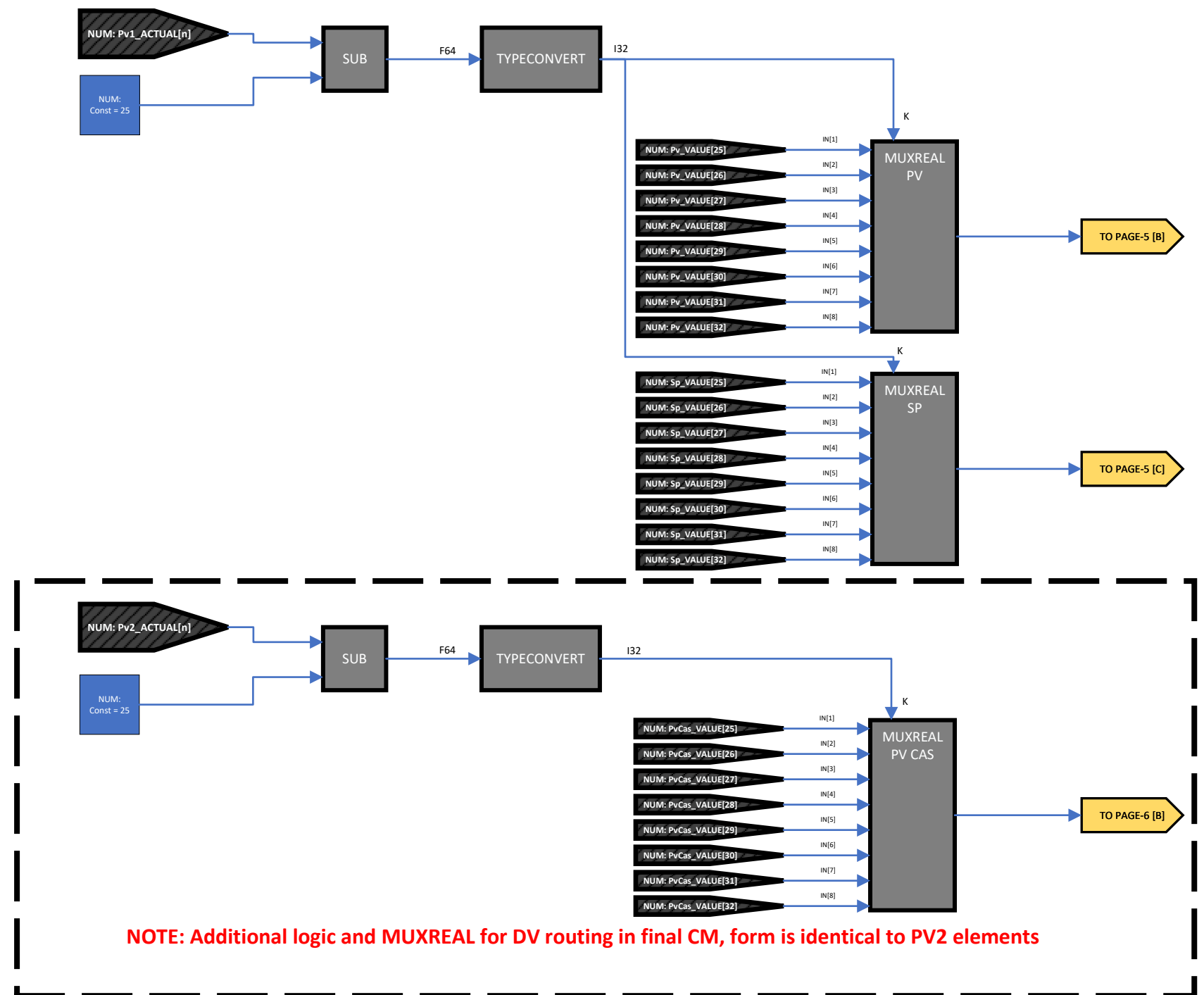
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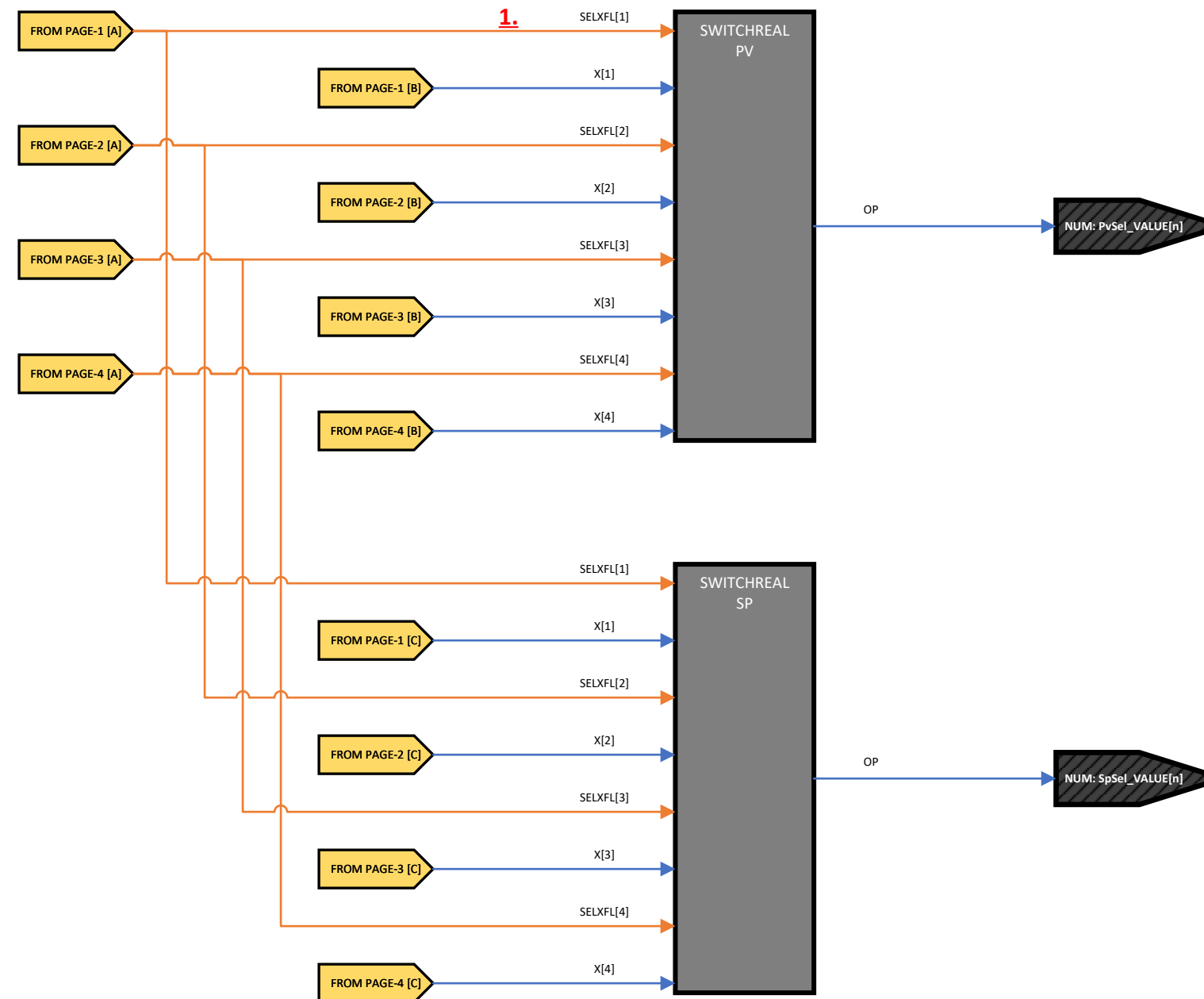
Page 2





Same as page-1, except checking desired SP, PV, and CAS PV values 25-32.

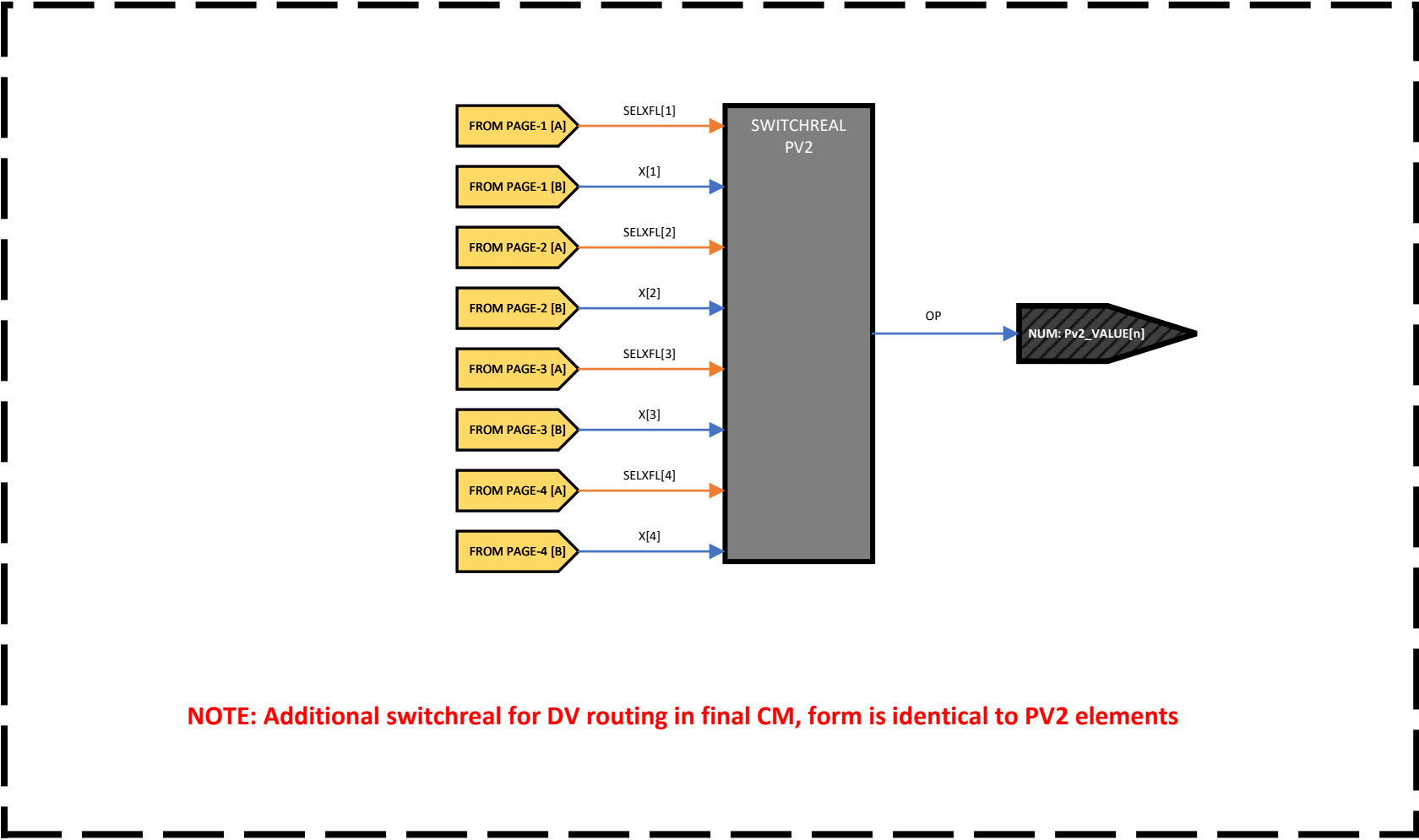
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1) a) If SELXFL[n] is true X[n] is routed to OP, if X[n], X[n+1], ..., X[n + m] are true, X[n] is routed to OP, i.e., the lowest value true flags corresponding input is routed to OP. This is fine as for this use case it is only possible for one of the flags, 1-4, to be true at any one time. As such no errors can take place in routing as a result of true flags input into the SWITCHREAL blocks (i.e., Equation C).

b) If no input flags are true, OP is written to by the last input to have had a true associated input flag, i.e., if all flags are false and last true flag was SELXFL[2], X[2] will be routed to OP. If program has only just been initialized, and no input flags have EVER been true, OP is written to by X[1]. Thus it is necessary to have a selectable PV and cascade PV corresponding to "NO PV SELECTED" that the switch will default to when no PVs are selected. This selection should correspond to a constant value of 0 routed to SWITCH OP.

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NOTES: <i>Same as page-5, except for routing selected CASCADE PV.</i>	Revision 0A 29-07-2022
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